

Resting-state EEG microstate dynamics in problematic media use: temporal expression, sequence syntax and overlap with emotional/internalizing symptoms in youth

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1. Background

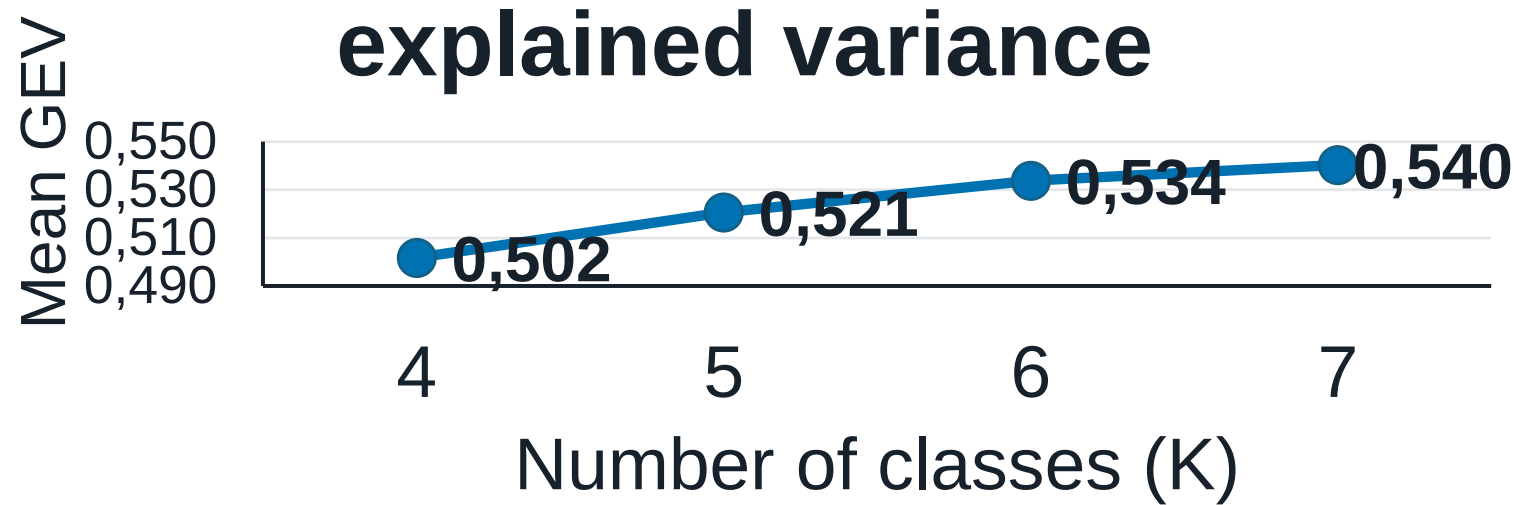
- Problematic media use (PMU) is increasingly relevant in child and adolescent mental health.
- An adult study linked problematic internet use to increased microstate E occurrence and coverage [1].
- No microstate study of PMU in youth is currently available

Aim: We tested associations between canonical microstate dynamics and sequence syntax with PMU and whether effects overlap with emotional/internalizing symptoms in a clinically relevant sample.

2. Methods

- Sample:** N = 40, age 10–19 years (M = 13.47). Rostock Res@t study site [2].
- Psychometrics:** Self-report PMU (GADIS-A, SOMEDIS-A, STREDIS-A [3]) and SDQ Emotional/Internalizing.
- EEG/Microstates:** 31-channel resting state, 1–30 Hz, ASR + ICA, interpolation, average reference. Median usable data: 299 s. Shared A–E templates and full-signal backfitting. Primary smoothing b = 30 ms [4].
- Sequence analysis:** fixed 200 s valid data segment, continuous and jump sequences
- First-order syntax:** expected-corrected jump ΔTM , 20 edges, PERMANOVA + edgewise follow-up models.
- Higher order organisation:** continuous AIF and jump ER (k = 4) versus individual first-order Markov surrogates [5]. AIF additionally vs. state-specific semi-Markov surrogates preserving Dwell-time dist.
- Covariates:** age, sex and medication

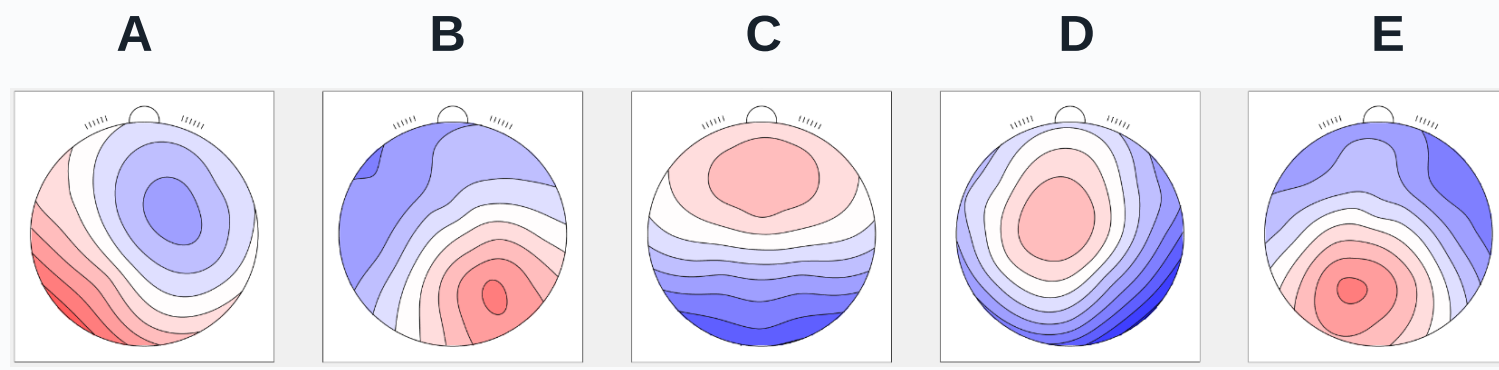
K selection: global explained variance



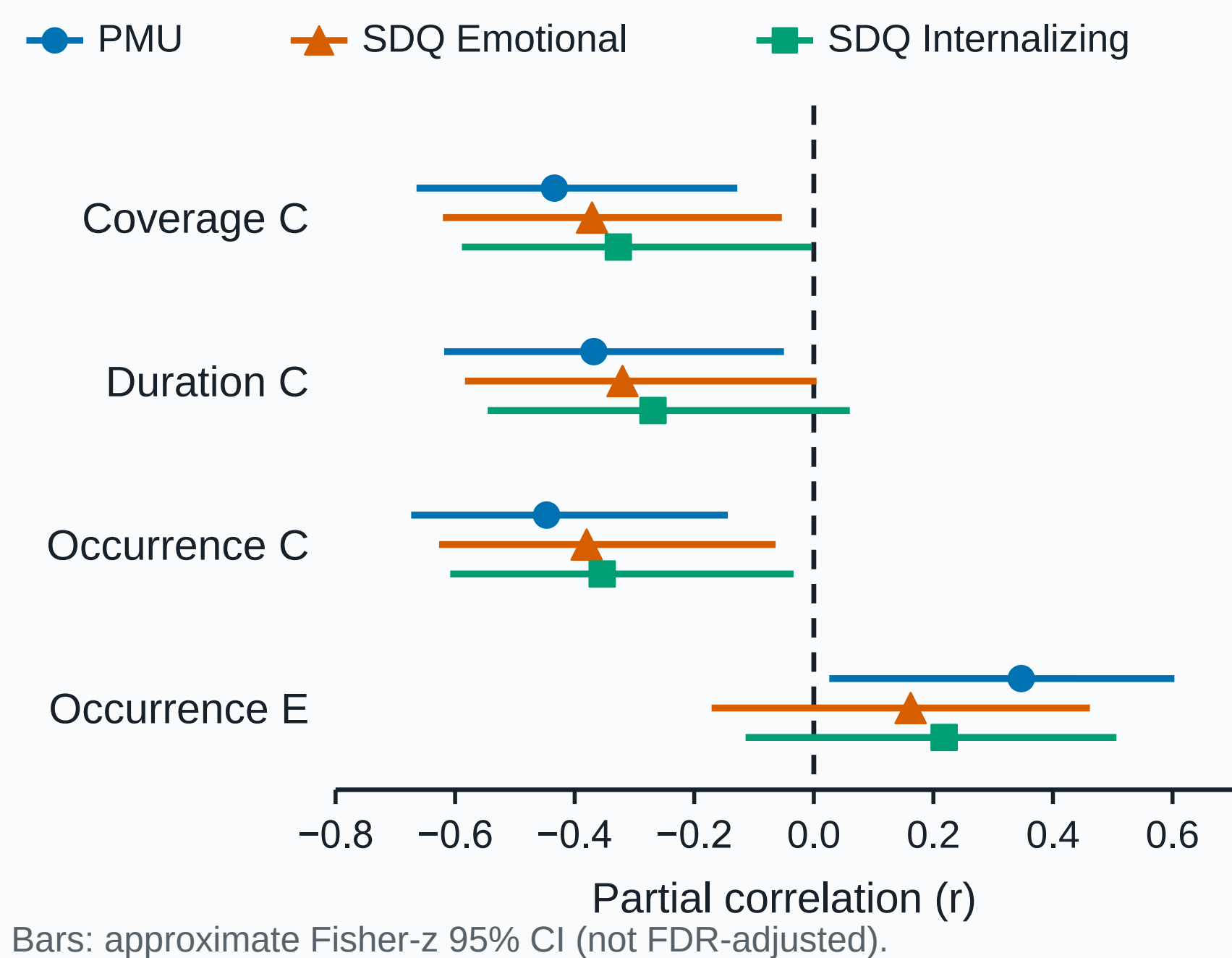
GEV gain: +1.89, +1.32 and +0.65 percentage points

K = 5 balances explained variance, parsimony and canonical interpretability.

Shared grand-mean templates (K = 5)



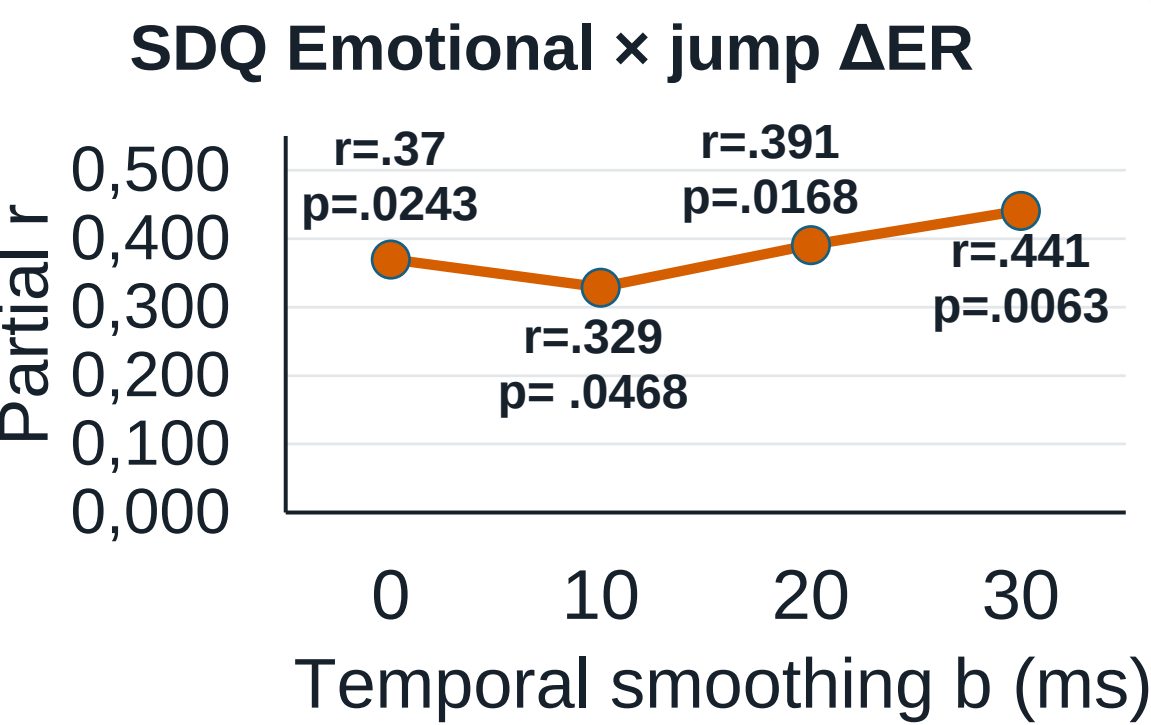
3. Results – Temporal expression



PMU effects overlap strongly with emotional and internalizing symptoms.

Reduced microstate C expression showed a similar pattern for PMU and SDQ emotional/internalizing symptoms, whereas increased E occurrence was comparatively more PMU-linked.

5. Results – Higher-order organization



Emotional x ΔER association: Persists after PMU adjustment at b = 30 (r = .378, p = .0231).

Continuous sequences: Ordinary Markov ΔAIF = -15,139 Semi-Markov ΔAIF = 0,725 -> Dwell times explain 95.4% (median) of Markov discrepancy at the b = 30 Dwell times matter.

Jump sequences: Group-level ΔER was smoothing-dependent and converged toward the first-order Markov expectation at b=30 (p = .256). Median ΔER -.0543 (b = 0) versus +.0027 (b = 30).

4. Results – First-order syntax

Expected-corrected jump transitions at b = 30 ms

		PMU					SDQ Emotional					SDQ Internalizing				
		Global R ² .051, p .0166					Global R ² .041, p .0756					Global R ² .039, p .1014				
		Next state					Next state					Next state				
Current state	A	B	C	D	E	A	B	C	D	E	A	B	C	D	E	
		.08	-.23	.04	.17		-.16	-.01	.22	.03		.01	.01	.05	-.03	
	B	.29		-.31	.04	.18	.12		-.32	.16	.27	.03		-.27	.07	.33
	C	-.06	-.23		.04	.23	.06	-.18		.08	.08	.08	-.18		.03	.09
	D	.21	-.02	-.35		.25	.10	.10	-.22		.08	-.10	.13	-.13		.09
E	-.13	.31	-.44	.34		-.04	.30	-.53	.28		.07	.35	-.59	.21		

Rows = current state, columns = next state. Cells show partial r, not ΔTM values. Blank diagonal = no self-transition. Black frames highlight E→C, not significance.

Global PMU association robust across smoothing

b (ms)	0	10	20	30
R ² / BH q	.0592 / .0464	.0511 / .0474	.0462 / .0464	.0511 / .0464

BH family: four smoothing settings. After SDQ Emotional adjustment: PMU R² = .0310, p = .2548. Global SDQ tests are not significant.

6. Interpretation

- PMU-related C and transition effects overlap substantially with SDQ symptoms; specificity remains unclear.
- First-order syntax adds information beyond conventional state metrics and is globally robust across smoothing.
- Continuous-sequence non-Markovianity is largely explained by dwell-time structure.
- Higher-order metrics are smoothing-dependent, but the Emotional- ΔER association persists across b.
- Higher emotional symptom burden was associated with ΔER values closer to the first-order Markov expectation.
- Limits: N=40, cross-sectional, shared SDQ and PMU self-report, exploratory analyses

7. Take-home message

- Higher PMU relates to reduced microstate C expression and increased E occurrence at baseline.**
- Global first-order transitions differ as a function of PMU, with substantial SDQ overlap. prominent : less E→C**
- More emotional problems are associated with fading higher order syntax.**

- [1] Simkute et al. Brain Topogr. 2025;38:68. doi:10.1007/s10548-025-01145-8
- [2] Labrenz et al. Front Psychiatry. 2025 Jan 20;15:1478012. doi: 10.3389/fpsy.2024.1478012.
- [3] (Somedis) Paschke K, Austermann MI, Thomasius R. J Clin Med. 2020;9:993. doi:10.3390/jcm9040993
- [4] Nagabhushan Kalburgi et al. Brain Topogr. 2024;37:621–645.
- [5] von Wegner et al. Brain Topogr. 2026;39:45. doi:10.1007/s10548-026-01196-5